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Rachel Hoffman-Bice

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EDUCATION

Massachusetts Institute of Technology , Cambridge, MA Doctor of Philosophy in Mechanical Engineering, Minor in Leadership and Management	GPA: 4.9/5.0	Expected June 2021
Massachusetts Institute of Technology , Cambridge, MA Master of Science in Mechanical Engineering	GPA: 4.9/5.0	June 2017
University of Florida , Gainesville, Florida Bachelor of Science in Mechanical Engineering, <i>summa cum laude</i> , <i>Outstanding Scholar Commencement Award</i> Minor in Electrical Engineering	GPA: 3.97/4.0	May 2015

RELEVANT PUBLICATIONS

- **R. Hoffman** and H. H. Asada, " Precision Assembly of Heavy Objects Suspended with Multiple Cables from a Crane," in *IEEE Robotics and Automation Letters*, Accepted June 2020.
- **Hoffman, R.**, and Asada, H., "A Multi-Track Elevator System for E-Commerce Fulfillment Centers", *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Vancouver, 2017. **Best Application Paper Award**
- Chacon-Castano, J., Rathbone, D., **Hoffman, R.**, et. al., "Music and the Brain – Design of an MEG Compatible Piano", *2017 International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2017) C*, Jeju Island, Korea

PROFESSIONAL EXPERIENCE

d'Arbeloff Laboratory, Massachusetts Institute of Technology **August 2015-Present**
Ph.D Research

- Investigated methods by which heavy shafts suspended with multiple cables at specific positions and orientations can be inserted into a chamfered hole despite a small clearance
- Analyzed key conditions for successful peg insertion by analyzing different cable attachment positions and orientations
- Developing a scaled prototype of a three-winch system with linear actuators designed to manipulate a 10 kg payload, including software and control of the system

Masters Research

- Invented a novel mechanism by which elevators can travel along tracks, and transfer between track segments of varying direction. This mechanism was inspired by a railway turnabout and the rack-pinion drive system of cog-railways.
- Analyzed optimal scheduling routes for multiple carts using this system and detailed the advantages over current elevator systems in e-commerce fulfillment centers.
- Designed and manufactured/implemented scale model consisting of thirteen direction changing mechanisms and twenty-seven track segments

MMID Full Service Design Team, Delft, The Netherlands, Functionality Design Intern **May 2017-Aug-2017**

- Extensive concept development of a high-pressure pipe coupling, an apparatus for testing chemical dilution and a versatile electric outlet wall box, through use of verbal and non-verbal communication, including detailed hand-sketched concepts.
- Performed mechanical tests on wall mounted clamp prototypes to ensure material selection and design analysis met client specifications

MIT Lincoln Laboratory, Systems and Analysis Group, Mechanical Engineering Intern **May 2014-Aug-2014**

- Evaluated and tested strategies to develop optimal terrain following aircraft trajectory method
- Designed and integrated terrain following MATLAB algorithm into radar testing software

Cummins Inc. Fuel Systems Group, Product Validation Intern **May 2013- August 2013**

- Analyzed the failure of a fuel filter and the implementation of preventative measures to ensure failure does not reoccur.
- Applied investigative problem solving methods to determine sources of failure for high pressure fuel lines, including resonant frequency and surface finish analysis.

CAMPUS INVOLVEMENT

Graduate Resident Tutor, Baker House, MIT **August 2016- Present**

- Foster an inclusive, respectful, supportive, safe, and positive living environment through mentoring, educating, encouraging personal growth, providing outlets for managing stress, and facilitating positive interpersonal relationships.

Medical Device Design Mentor, Department of Mechanical Engineering, MIT **August 2017- Present**

Rachel Hoffman

- Provide extensive mechanical design consulting for student-lead medical device design teams, including the development of a Crank-Powered Otoscope, a Center-of-Gravity Detector for Heavy Machinery Operator Fatigue, a Lower Extremity Fracture Fixation Device, a Preterm Vaginal Breech Delivery Device and a Portable Device for helping the physically impaired stand

RELEVANT COURSEWORK AND SKILLSET Medical Device Design, Precision Machine Design, Deterministic and Exact Constraint Design, MATLAB, Python, ROS, Solidworks